

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2: SYMMETRY AND RELATIVITY

TRINITY TERM 2024

Wednesday, 5 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) The Cosmic Microwave Background radiation is a relic of the big-bang filling the universe with photons of average energy 6.626×10^{-4} eV. Ultra-high energy photons can collide with these. Find the minimum energy one of these photons must have to produce an electron positron pair (the mass of a positron and electron are equal). [5]

(b) The helicity of a fundamental particle is defined as the scalar product of the direction of its spin with the direction of its momentum $\hat{s} \cdot \hat{p}$ (take spin to be a direction - do not use spinors in this problem). An electron of finite mass M_e and momentum of magnitude p has helicity +1. Specify the boost parameter ($\vec{\beta}$) for a frame where the helicity is -1 in terms of the electron's energy and momentum. A massless particle has helicity +1; is there a frame in which its helicity is -1? Give a very short explanation. [3]

(c) Consider the expression $m \frac{d^2 x^\alpha}{d\tau^2} = q F^{\alpha\beta} U_\beta$ for the 4-force on a particle of charge q , where U_β is its 4-velocity and $F^{\alpha\beta}$ is the Maxwell field tensor. Show that replacing $d\tau$ with $-d\tau$ changes this to an equation of a particle of opposite charge - why is this true in any frame? A Lorentz scalar needed to describe the collision of two electrons is

$$\frac{(p_2^f \cdot p_1^f)(p_1^i \cdot p_2^i) + (p_2^f \cdot p_2^i)(p_1^i \cdot p_1^f)}{(p_2^f - p_1^i)^4},$$

where subscripts 1, 2, refer to the electrons and i, f label initial (before the collision) and final (after the collision) 4-momenta respectively. Convert this to an expression for a positron (same mass but opposite charge to an electron) in place of electron 2, where the positron has initial 4-momentum $= p_{e+}^i$, and final 4-momentum $= p_{e+}^f$ (think about the order of events that an observer with a normal sense of time sees in contrast to the positron's viewpoint). [5]

(d) A photon with 4-momentum $(\frac{E_1}{c}, \vec{p}_1)$ collides with an electron at rest and is scattered at an angle θ to its original trajectory. The electron now has 4 momentum $(\frac{E_e}{c}, \vec{p}_e)$ and the scattered photon has 4-momentum $(\frac{E_2}{c}, \vec{p}_2)$. Show that the difference in wavelengths of the photons is equal to $\frac{h}{M_e c} (1 - \cos \theta)$ (where h is Planck's constant). Which photon (1 or 2) has the shorter wavelength and why? [6]

(e) Convert the expression in part (d) to the corresponding expression for a positron at rest colliding with an electron $(\frac{E_e}{c}, \vec{p}_e)$ and annihilating into two photons. Use your result to show that for electron positron annihilation into two photons $\lambda_1 + \lambda_2 = \frac{h}{M_e c} (1 - \cos \theta)$, where θ is the angle between the two emerging photons and λ_1 and λ_2 are their wavelengths. [6]

2. (a) Consider the equation $m \frac{dU^\alpha}{d\tau} = q F^{\alpha\beta} U_\beta$ for the motion of a charged particle where U^α is the 4-velocity $= (c\gamma, \gamma\vec{v})$, and $F^{\alpha\beta}$ is the Maxwell field tensor. Consider the case where only a magnetic field is present, and show that the 3-force acting on the particle is orthogonal to its spatial velocity. Find the radius of curvature of the track in terms of the B field, γ , the particle's mass m , its charge q and its speed v - justify any assumptions. [5]

(b) Consider the electromagnetic (EM) 4-potential $A^\mu(x) = (\frac{\phi(x)}{c}, \vec{A}(x))$ which satisfies the equation $\partial_\nu \partial^\nu A^\mu - \partial^\mu \partial_\nu A^\nu = -\mu_0 J^\mu(x)$, where $J^0 = c\rho$. For $\vec{A} = 0$ and a time independent A^0 , recover Poisson's equation $\nabla^2 \phi = -\frac{\rho}{\epsilon_0}$. Do not make any further gauge choices. State the other components of J^μ for this particular case (here μ_0 is the magnetic permeability of the vacuum, c is the speed of light and ρ is the charge density, $\phi(x)$ and $\vec{A}(x)$ are the electro-magnetic scalar and vector potentials respectively). [5]

(c) Again do not assume any particular gauge. Consider a 4-vector potential $A^\mu(x) = \epsilon^\mu \sin(k^\alpha x_\alpha)$ where ϵ^μ is a unit 4-vector and k^α is a 4-vector. The Maxwell field tensor is $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. Show that the components $F^{0i} = -\frac{E_i}{c}$ and $F^{ij} = \epsilon_{ijk} B_k$ where i and j are spatial indices and E_i B_j are the components of electric and magnetic fields. What is $\partial_\mu F^{\mu\nu}$ for a source free region of space and why? [5]

(d) Work out and write down explicitly the elements (in terms of E and B) of the tensor $\tilde{F}_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} F^{\alpha\beta}$, where $\epsilon_{\mu\nu\alpha\beta}$ is the totally anti-symmetric Levi-Cevita tensor density in 4-dimensions. For the potential in (c), compute the invariants $\epsilon_{\mu\nu\alpha\beta} F^{\alpha\beta} F^{\mu\nu}$ and $F_{\mu\nu} F^{\mu\nu}$ in terms of the components of k and ϵ and hence show that the E and B components of the electromagnetic wave are orthogonal and $|\vec{B}| = \frac{|\vec{E}|}{c}$ without assuming any particular gauge. [8]

(e) An observer measures electric and magnetic fields in one frame to be $(E, E, 0)$ and $(0, 0, \frac{2E}{c})$ respectively, whereas another observer moving relative to them measures $(0, 0, E)$ and $(B_1, \frac{E}{c}, B_2)$. Find B_1 and B_2 - state what invariants you are using. [2]

3. (a) A triangle of sides $3L$, $4L$ and $5L$ lies with its shortest side along the x' -axis in frame S' which moves to the right relative to frame S (standard configuration), with velocity v . What are each of the three sides of the triangle as measured in frame S and what is the area of the triangle in S' ? What is the angle the longest side makes with the x axis? [3]

(b) Consider a wave train of monochromatic light in frame S moving in an arbitrary direction in the x - y plane and consider the distance L between two points on the same wavefront. Take the wave front to be infinitely long on either side. Show that L^2 is a Lorentz Invariant (choose the moving frame S' to coincide with S at the origin). (Hint: place one point at the origin and trace the location of the other point in S and S' , bearing in mind all the consequences of the finite speed of light). [10]

(c) Show that the ratio of intensities of the wave (energy per unit area-time) in two frames is $(\frac{\omega'}{\omega})^2$, where ω , ω' are the angular frequencies as measured in the two frames. (Hint: The intensity varies as the number of photons per volume of area oriented in the direction of propagation and length that can be taken as the wavelength.) [5]

(d) A moving source emitting light of frequency ω isotropically in its own rest frame approaches an observer with boost parameter β and $\gamma = \frac{1}{\sqrt{1-\beta^2}}$. Given that $\omega = \gamma\omega' - \gamma\beta\omega' \cos \theta'$, where the primed frame is the observer's frame, write down $\cos \theta$, where θ is the emission angle in the source's rest frame, in terms of γ , β and $\cos \theta'$ and hence show that $\frac{d\Omega}{d\Omega'} = (\frac{\omega}{\omega'})^2$ where $d\Omega$ and $d\Omega'$ are the differential solid angle elements along the line of the source's trajectory. [5]

(e) Using the results of parts (c) and (d), show that $\frac{\frac{dP}{d\Omega'}}{\frac{dP}{d\Omega}} = (\frac{\omega'}{\omega})^4$, where $\frac{dP}{d\Omega}$ is the power emitted by the source per differential solid angle. [2]

4. (a) A particle of mass m moves from rest and from the origin under the influence of a constant force F_0 (3-force) for a time t . Show that the relativistic formula for its speed for motion in one dimension is $\frac{F_0 t}{m\sqrt{1+\frac{F_0^2 t^2}{m^2 c^2}}}$ and calculate its displacement from the

origin in terms of F_0 , m , t , and the speed of light c . Show that it's motion is parabolic with time in the non-relativistic limit. [5]

(b) A spaceship's world line is given by $x^2 - c^2 t^2 = \frac{c^4}{a_0^2}$. Show that a_0 is the acceleration in its instantaneous rest frame. [5]

(c) Show that the world line in part (b) can be rewritten as $\cosh^2 \frac{a_0 \tau}{c} - \sinh^2 \frac{a_0 \tau}{c} = 1$: first choose $\sinh \omega = \frac{a_0 t}{c}$ and show that $v(t) = c \tanh \omega$, then use $\gamma = \frac{dt}{d\tau}$, where τ is the proper time (you may need $\cosh^2 A - \sinh^2 A = 1$, $\frac{\sinh A}{\cosh A} = \tanh A$). [5]

(d) A spaceship experiences constant acceleration in its own rest frame and an observer B at rest sees its world line is $x^2 - c^2 t^2 = \frac{c^4}{a_0^2}$. At time $t = -\frac{D}{c}$ (where D is a positive constant), B fires a photon at the spaceship from $x = 0$. Find the proper time in the spaceship at which the photon reaches it. [5]

(e) Referring to the spaceship in part (d), show algebraically that a photon fired at the spaceship at time $t > 0$ will never reach it. To convince yourself, you may want to sketch a space time diagram of all trajectories in part (d). This is an example of the Rindler Horizon. [5]